

LUBRICANTS

Introduction.

In all types of machines, the surfaces of moving, sliding or rolling parts run against each other. The mutual rubbing of one part against another generates frictional force which offers resistance to the relative motion of these surfaces. Wear results when this resistance is overcome by applied forces. Hence friction causes a lot of wear & tear of surfaces which consequently require repeated replacement. It causes loss of efficiency of machine.

These drawbacks can be minimized by applying a thin layer of certain substance, known as lubricant in between moving/sliding/rolling surfaces. A lubricant may ~~there~~ be defined as a substance which reduces the friction when introduced between two surfaces and the phenomenon is known as lubrication.

Functions of lubricants :

- i) The first and foremost function is to reduce friction.
- ii) It reduces wear, tear and surface deformation.
- iii) It acts as coolant to carry away heat.
- iv) It keeps out dirt.
- v) Prevents rust and corrosion.
- vi) It transmits fluid power.

Used as cutting fluid in cutting, grinding & drilling of ^{metals.} ~~machines~~

Liquid lubricants are used in medicine.

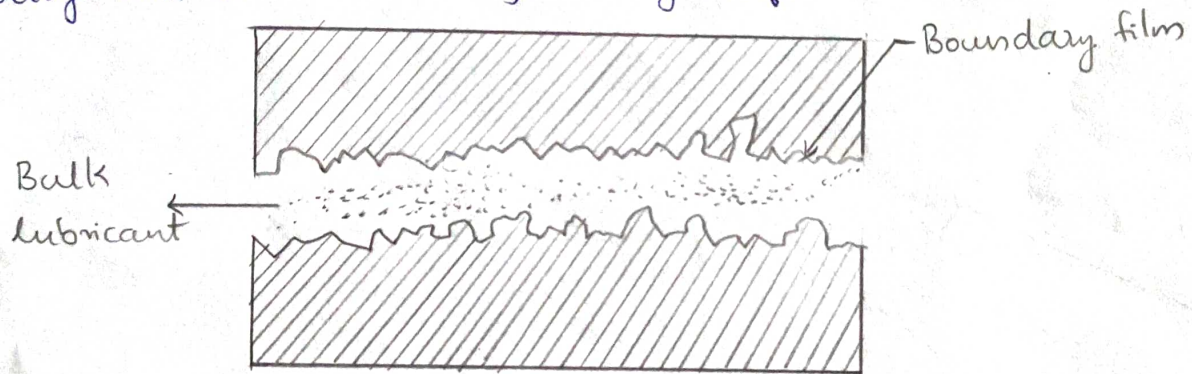
Used in paints and soaps.

MECHANISM OF LUBRICATION

Lubrication mechanisms can be classified into following:

a) Hydrodynamic lubrication / Thick film / Fluid film lubrication.

In this moving/sliding surfaces are separated from each other by a bulk lubricant film. This prevents direct surface to surface contact so that small peaks and valleys do not interlock. This consequently reduces friction and prevents wear. The small friction is only due to internal resistance between particles of lubricant moving over each other. In such a system, friction depends on thickness and viscosity of lubricant and relative velocity and area of moving/sliding surface.

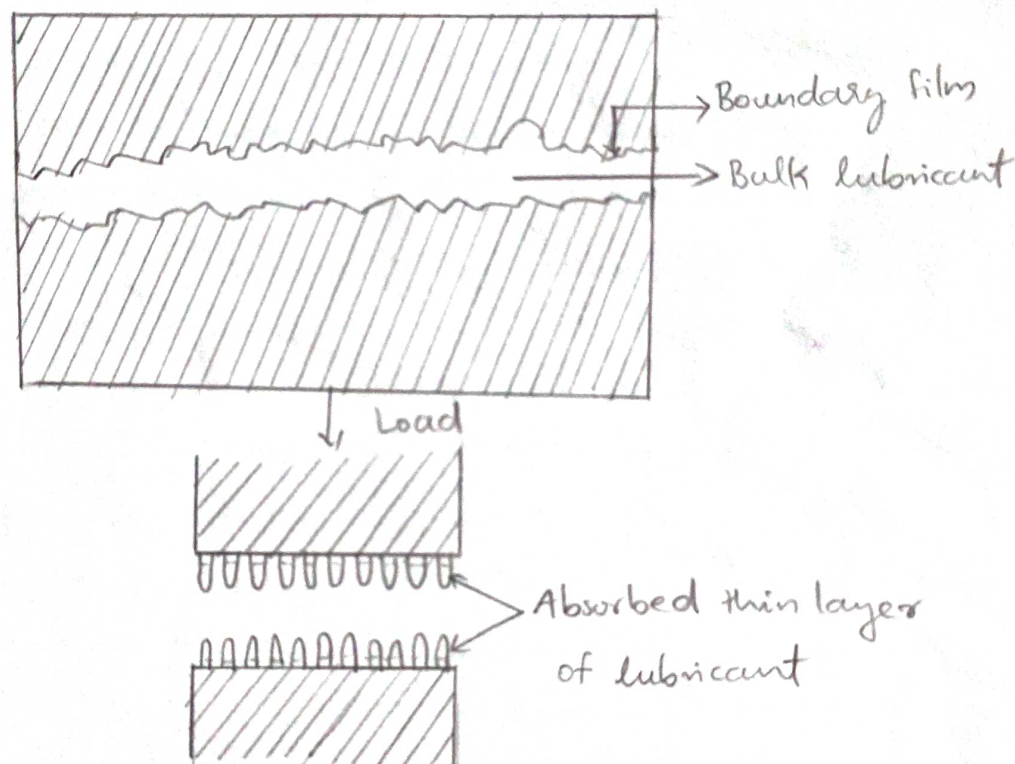


Fluid film lubrication is statistically done by hydrocarbon oils. These are generally blended with selected long chain polymers in order to maintain viscosity of oil constant in all seasons of year. This is necessary because viscosity of hydrocarbon oil decreases with rise in temperature.

b) Boundary lubrication (thin-film lubrication)

When the lubricant is not viscous enough to generate a film of sufficient thickness to separate the surfaces under heavy loads, friction may yet be reduced with proper lubricant. Such an application is boundary lubricant. A thin layer of lubricant is adsorbed on the metallic surfaces which avoids

direct metal to metal contact. The load is carried by the layer of adsorbed lubricant on both metal surfaces. The distance between moving surface is very small of the order of the height of surface asperities. Contact between metal surface is possible by the squeezing out of lubricating oil film. When this occurs the load would be taken on the high spots of journal and bearing and two surfaces tend to become welded together by appreciable heat generated. This prevents motion as the 2 surfaces adhere together. This is "seizure"



For boundary lubrication lubricant molecules should be:

- i) Long hydrocarbon chains
- ii) Lateral attraction b/w the chains
- iii) Good oiliness
- iv) Low pour and oxidation
- v) High viscosity index
- vi) Polar groups to promote wetting or spreading over the surface.

Mechanism of lubrication.

In order to lubricate, we have to consider the conditions under which the machine is working.

- a). Thick^{film}/fluid/hydrodynamic lubrication.
- b). Thin film/boundary lubrication.
 - Fluid film lub. useful in delicate & light machines like watch, clock, guns, scientific equipments.
 - Thin film used at low speed & heavy load (Pressure)
- c). Extreme pressure/Temperature lubrication.
 - Surfaces are under high pressure & speed.
 - High temp. generated due to friction.
 - liquid lub fail to stick, decomposes & vapourizes.
 - additives + mineral oil_{spcl}
 - Organic compound having groups like chloride, sulphur, P.
 - They react with metallic surfaces to form metallic compounds.

SOLID LUBRICANTS

Solid lubricants are materials, which in solid phase reduce friction between surfaces sliding against each other, without the need for a liquid medium. These are used as powder or thin film. Other terms commonly used for solid lubrication include dry lubrication, dry-film lubrication, and solid film lubrication. Although these terms imply that solid lubrication takes place under dry conditions, fluids are frequently used as a medium or as a lubricant with solid additives.

They are able to provide improved lubrication at higher temperatures compared to traditional oil-based lubricants. Low-friction properties of dry lubricants are attributed to their layered structure on the molecular level with weak bonding between layers. Such layers are able to slide relative to each other with minimal applied force, therefore giving them their low friction properties. However, a layered crystal structure alone is not necessarily sufficient for lubrication. There are some solids with non-lamellar structures that function well as dry lubricants in certain applications such as polytetrafluoroethylene (PTFE or teflon). The most commonly used solid lubricants are graphite, molybdenum disulphide (MoS_2), PTFE and hexagonal boron nitride (hBN).

Graphite

Graphite is a crystalline, low density and soft allotrope of carbon. It is a solid lubricant relating to the class of inorganic lubricants with lamellar structure. The crystal lattice of graphite consists of hexagonal rings forming thin parallel planes bonded to three other atoms in the plane. The graphenes are bonded to each other by weak Van der Waals forces.

The layered structure allows sliding movement of the parallel planes. Weak bonding between the planes provides low shear strength in the direction of the sliding movement but high compression strength in the direction perpendicular to the sliding movement. Friction forces cause the graphite particles to orient in the direction, in which the graphenes are parallel to the sliding movement. The anisotropy of the mechanical properties imparts the combination of low coefficient of friction and high carrying capacity to graphite.

Graphite forms a lubrication film strongly adhered to the substrate surface. The lubrication film provides good wear resistance and seizure resistance (compatibility). Lubricating properties of graphite are highly dependent on the presence of water vapour in the ambient atmosphere. Water molecules are absorbed on the graphite surface causing further reduction of the bonding between the graphene planes. It will work upto a temperature limit of 788°C .

Molybdenum disulphide.

The crystal lattice of molybdenum disulphide is similar to that of graphite. It consists of hexagonal molybdenum planes sandwiched between two hexagonal sulphur planes. The atoms in the planes are strongly attached to each other by covalent bond. Planes are bonded by weak Van der Waals forces.

The sulphur layers of molybdenum disulphide have an affinity for tenacious adherence to the metal substrate atoms therefore a strong lubrication film is formed on the substrate surface. The lubrication film provides good wear resistance and seizure resistance. In contrast to graphite, moist atmosphere is not required for lubrication by molybdenum disulphide. Therefore it demonstrates low friction in dry atmosphere and in vacuum where its coefficient of friction is even lower than in the presence of water vapour. Coefficient of friction of molybdenum disulphide is lower than that of graphite & decreases with .

increasing load.

PTFE

PTFE coating consists of carbon and fluorine atoms and is one of the most slippery materials since it has very low surface tension. Unlike other solid lubricants, PTFE does not have a layered structure. The molecules of PTFE slip along with each other very easily, similar to other lamellar structures. PTFE can be used upto a temperature of 260°C .

Advantages of solid lubricants.

- More efficient than fluid lubricants at high loads and speeds.
- High resistance to deterioration in storage.
- High stable in extreme temperature, pressure, radiation and reactive environments.
- Permit equipment to be lighter and simpler because lubrication distribution systems and seals are not required.